IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

Khaled HAOUAS
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# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

## Summary of Methodologies

1. Data Collection: Accessed SpaceX launch data via API and web scraped records from Wikipedia.

2. Data Cleaning & Preparation:

- Cleaned and formatted the data.
- Stored data in Db2 database and performed SQL queries.
- Conducted exploratory data analysis.

3. Feature Engineering: Created new features and standardized the data.

4. Interactive Visualizations:

- Mapped launch sites and success rates using Folium.
- Built an interactive dashboard with Plotly Dash.

5. Model Building & Evaluation:

- Implemented SVM, Decision Trees, and K-Nearest Neighbors.
- Tuned hyperparameters with GridSearchCV.
- Evaluated models using test data accuracy.

## Summary of Results

1. Data Insights:

- Identified factors influencing Falcon 9 first stage landings.
- Visualized geographical patterns and success rates.

2. Model Performance:

- SVM and K-Nearest Neighbors: 83.33% accuracy.
- Decision Tree: 94.44% accuracy.

3. Key Findings:

- Launch site and payload mass impact landing success.
- Decision Tree model is the most effective predictor.

# Introduction

**Project Background & Context**
For this capstone project, the objective is simple: predict whether the Falcon 9 first stage will land successfully.

One of the main reasons SpaceX can offer launches at a lower cost than competitors is because they reuse the first-stage booster. If the landing succeeds, the booster can fly again — which significantly reduces overall launch costs.

If we can reliably predict landing success, we can indirectly estimate launch cost efficiency. That kind of insight becomes especially useful for companies competing or bidding against SpaceX.

**Key Questions We're Trying to Answer**
- What variables impact the success of a Falcon 9 first-stage landing?
- Can we build a machine learning model that predicts the landing outcome with solid accuracy?
- Among the models we test, which one performs best and why?

Section 1

# Methodology

# Methodology

**Executive Summary:** This project employs a comprehensive approach to predict the successful landing of the Falcon 9 first stage, incorporating data collection, processing, exploratory analysis, interactive visualizations, and predictive modeling.

**Data Collection Methodology**: Data was sourced from the SpaceX API, which provided detailed records of Falcon 9 launches, including launch dates, sites, payloads, and outcomes.

**Perform Data Wrangling**: Data cleaning involved handling missing values, standardizing formats, and ensuring consistency. Key features were extracted and new features engineered to enrich the dataset.

**Perform Exploratory Data Analysis (EDA) Using Visualization and SQL:**

- Visualized launch success rates, payloads, and launch sites using Matplotlib and Seaborn.
- Executed SQL queries to derive insights and answer specific questions regarding the dataset.

# Methodology

**Perform Interactive Visual Analytics Using Folium and Plotly Dash:**

- Used Folium to create interactive maps displaying launch sites and outcomes.
- Developed a Plotly Dash application with interactive components like dropdowns and sliders to analyze launch success rates and payload ranges.

**Perform Predictive Analysis Using Classification Models:**

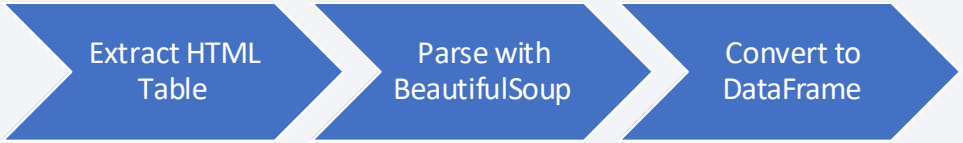
- Built and evaluated various classification models including Logistic Regression, SVM, KNN, and Decision Trees.
- Employed GridSearchCV for hyperparameter tuning.
- Evaluated models based on accuracy, and identified the best performing model for predicting landing success.

**Github URL:** https://github.com/Khaledhaouas/IBM-Data-Science-and-Machine-Learning-Capstone-Project

# Data Collection

- Step 1: SpaceX API Request

- Step 2: Web Scraping Wikipedia




- Step 3: Data Integration

# Data Collection - SpaceX API

**Step 1: Initiate API Request**

- Use Python's `requests` library to connect to the SpaceX API.
- Endpoint: `https://api.spacexdata.com/v4/launches`

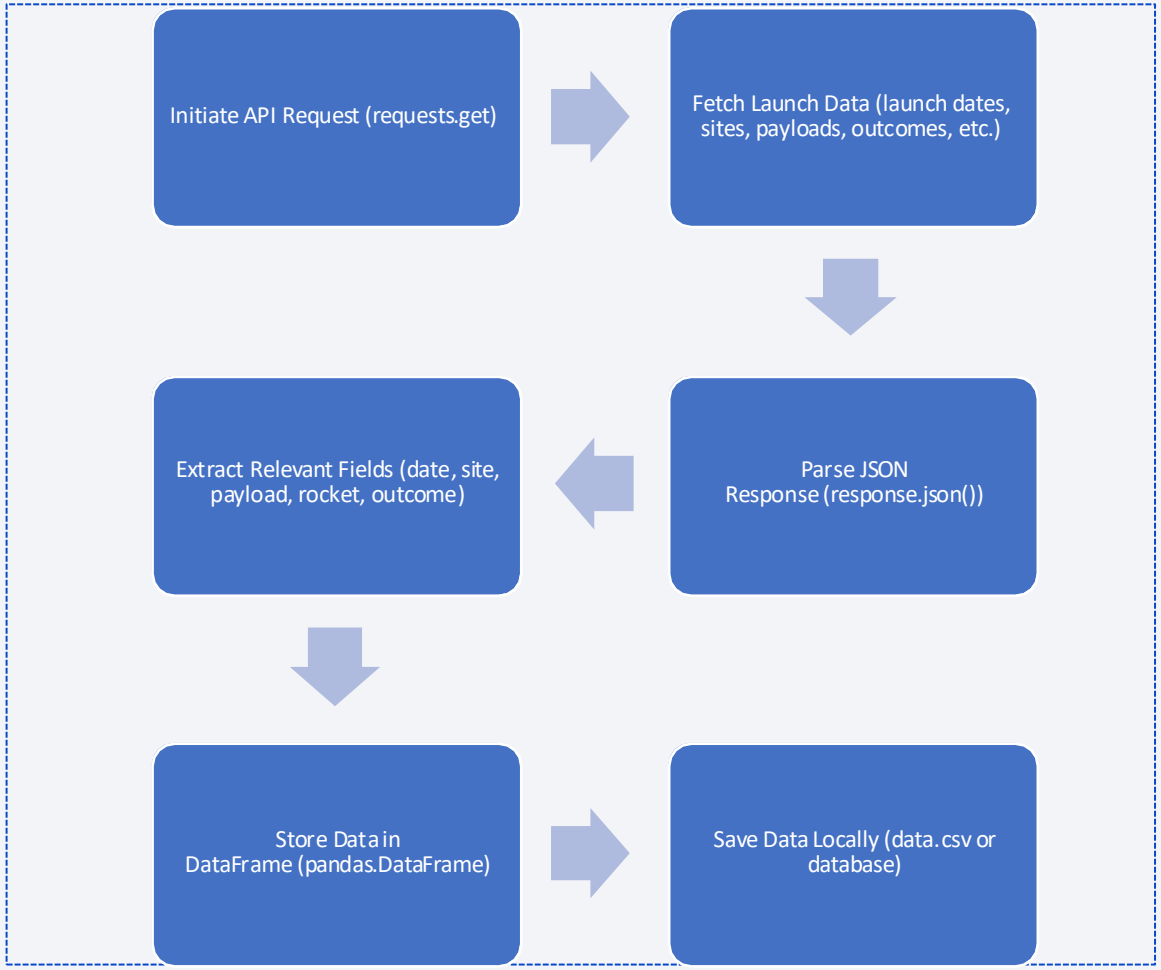
**Step 2: Parse API Response**

- Convert API response from JSON to a Python dictionary.
- Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.

**Step 3: Store Data Locally**

- Save extracted data into a pandas DataFrame.
- Store the DataFrame locally for further processing.

GitHub URL: 1. jupyter-labs-spacex-data-collection-api.ipynb



# Data Collection - Scraping

Step 1: Initiate Web Scraping

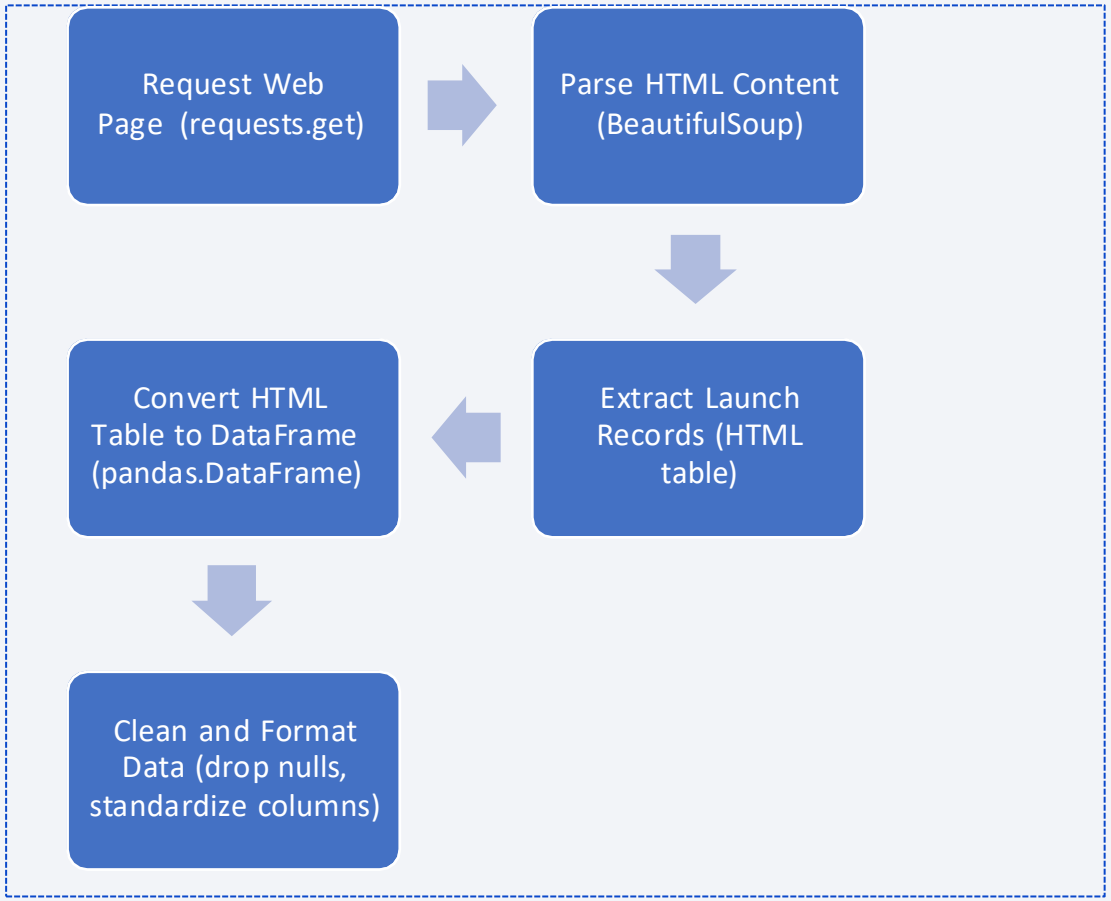
- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL: `https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches`

Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9 launch records.

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.




GitHub URL: 2. jupyter-labs-webscraping.ipynb

# Data Wrangling

**Overview:** Data wrangling involves cleaning, transforming, and organizing raw data into a structured format suitable for analysis.

- Step 1: Data Cleaning
  - Identify and fill or remove missing values in the dataset.
  - Use appropriate imputation techniques or drop rows/columns with excessive missing data.

- Step 2: Data Transformation
  - Convert data types to appropriate formats (e.g., date-time, numerical).
  - Standardize text (e.g., lowercase, remove whitespace).
  - Create new features from existing data (e.g., extract year from date).
  - Normalize/scale numerical features to ensure consistency.

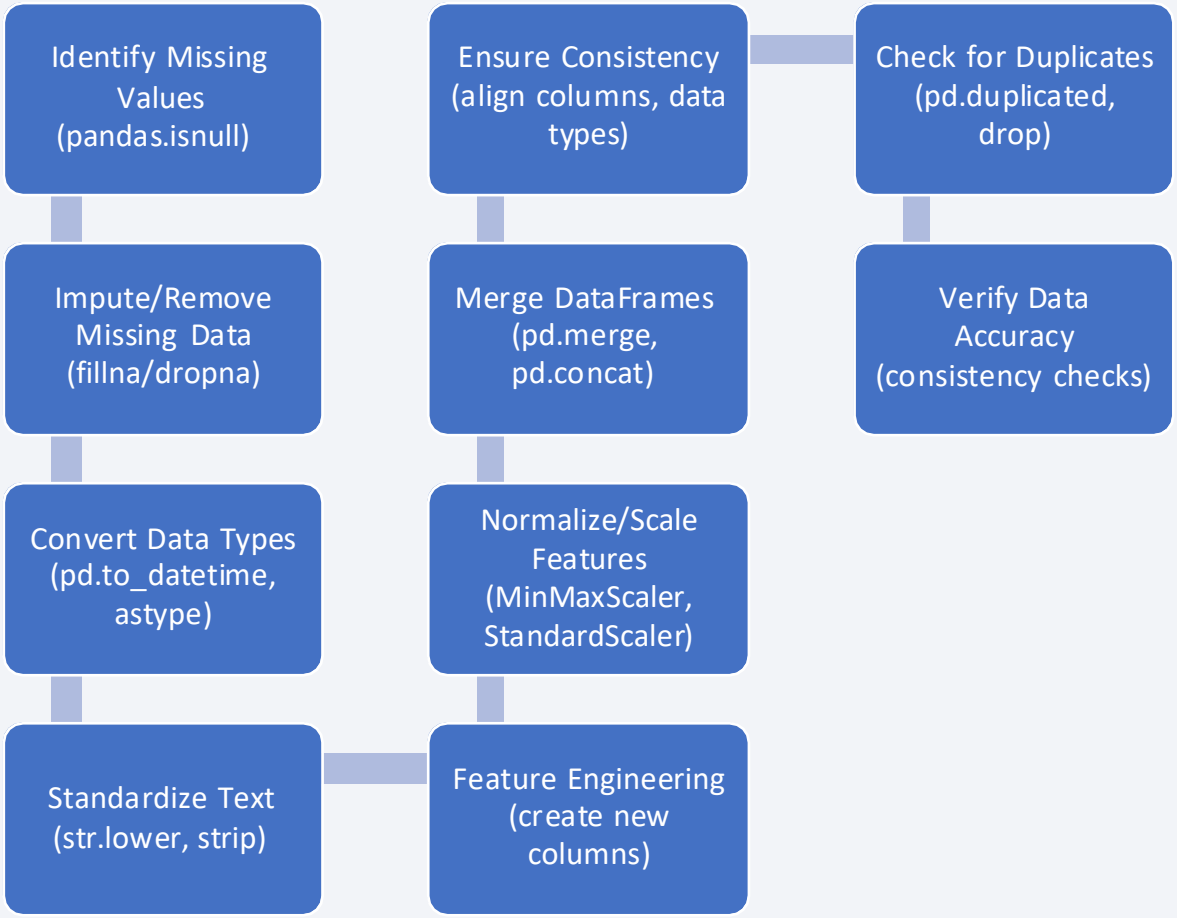
# Data Wrangling

- Step 3: Data Integration
  - Merge datasets collected from different sources (API, web scraping) into a single cohesive dataset.
  - Ensure consistent column names and data formats across datasets.

- Step 4: Data Validation
  - Check for duplicate records and remove them.
  - Verify the accuracy and consistency of data entries.

GitHub URL: 3. labs-jupyter-spacex-Data wrangling.ipynb

# Data Wrangling Flowchart



# EDA with Data Visualization

**Overview:**

Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

**Charts Plotted:**

**1. Histograms:**

- **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
  - **Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

**2. Bar Charts:**

- **Purpose**: Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
  - Why: Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

**3. Line Charts:**

- **Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- **Why**: Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

# EDA with Data Visualization

**4. Scatter Plots:**

- **Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.
- Why: Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.

**5. Heatmaps:**

- Purpose: Used to visualize correlation matrices between multiple numerical variables.
- Why: Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

**6. Box Plots:**

- Purpose: Used to display the distribution of numerical data through their quartiles.
- Why: Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

**Github URL:** 5. jupyter-labs-eda-dataviz.ipynb

# EDA with SQL

**Aggregate Queries:**

- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

**Join Queries:**

- Joined tables to link launch records with additional data (e.g., rocket details).
- Combined datasets for comprehensive analysis.

**Filtering Queries:**

- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

**Sorting Queries:**

- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

**Subqueries:**

- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

GitHub URL: 4. jupyter-labs-eda-sql-coursera_sqllite.ipynb

# Build an Interactive Map with Folium

## Map Objects Created

**Markers:**

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

**Circles:**

- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

**Lines:**

- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between different points of interest related to launches.

Github URL: 6. lab_jupyter_launch_site_location.ipynb

## Reasons for Adding Objects

**Markers:**

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

**Circles:**

- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

**Lines:**

- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

# Build a Dashboard with Plotly Dash

## Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Github URL: spacex_dash_app.py

## Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

# Reasons for Adding Plots and Interactions

**Success Pie Chart:**

- Provides a quick overview of mission success rates.
- Essential for stakeholders to understand overall performance metrics at a glance.

**Success-Payload Scatter Plot:**

- Helps identify correlations between payload characteristics and launch outcomes.
- Supports decision-making processes related to payload planning and operational strategies.

**Launch Site Dropdown:**

- Enhances user experience by focusing analysis on specific launch locations.
- Allows for regional insights and comparisons across different launch sites.

**Range Slider for Payload:**

- Offers interactive exploration of how payload mass affects mission success.
- Enables detailed analysis and insights into payload-related performance factors.

# Predictive Analysis (Classification)

1. Data Preprocessing:

- Standardized features to ensure all variables contribute equally.
- Split data into training and test sets for model validation.

2. Model Selection:

- Explored multiple classification algorithms: SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Chose algorithms suitable for binary classification tasks based on project requirements.

3. Hyperparameter Tuning:

- Used GridSearchCV to systematically search for optimal hyperparameters.
- Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

Github URL:7. SpaceX_Machine_Learning_Prediction_Part_5.jupyterlite.ipynb

4. Model Evaluation:

- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

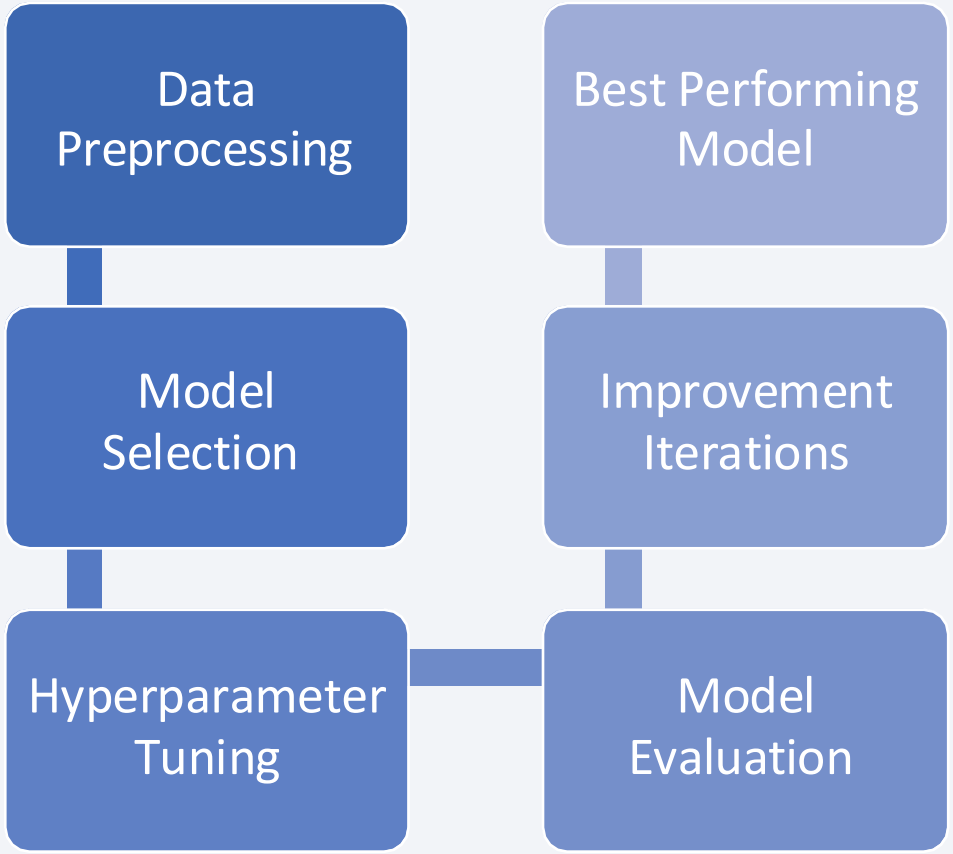
5. Improvement Iterations:

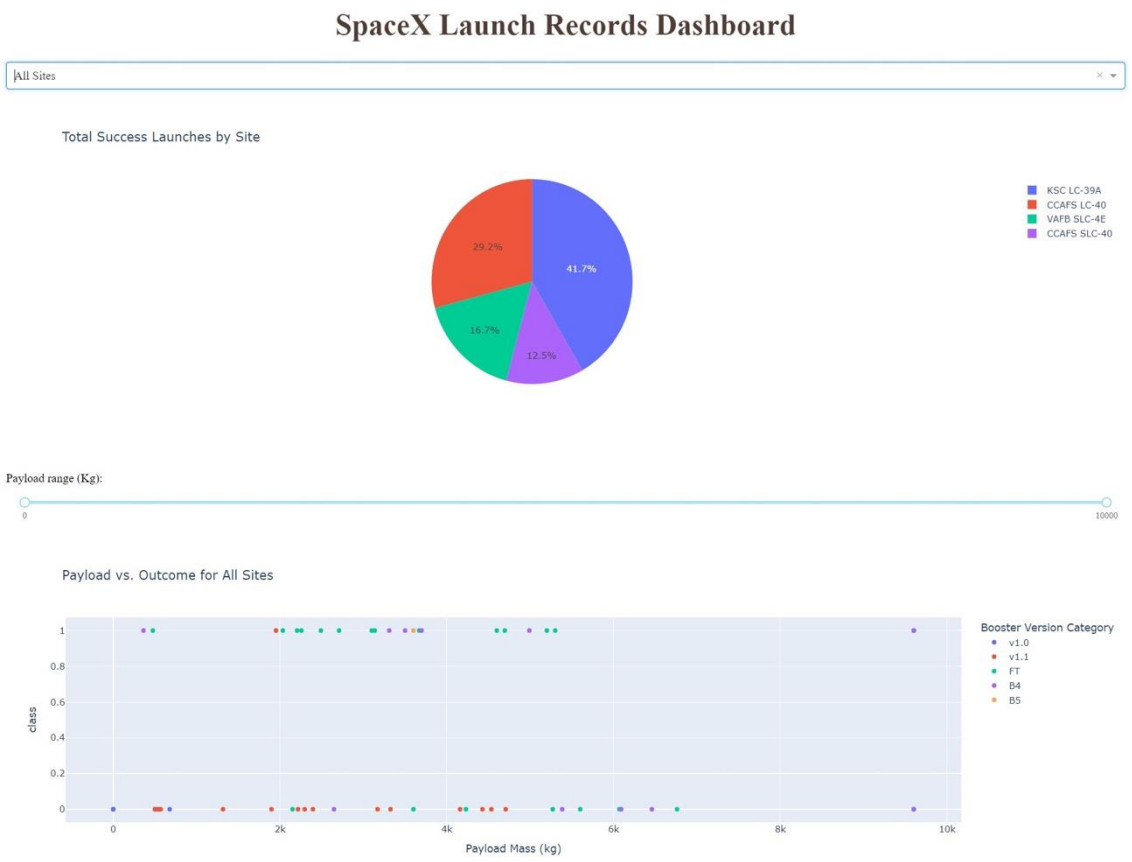
- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

6. Selection of Best Performing Model:

- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability.

# Predictive Analysis (Flowchart)

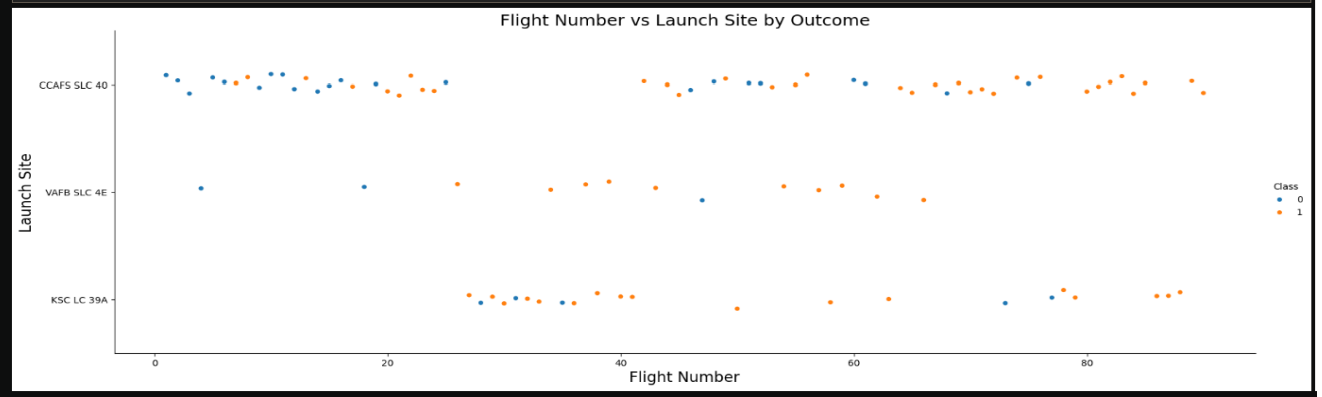





## TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`,set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

[10]:
```
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(x="FlightNumber", y="LaunchSite", hue="Class", data=df, aspect=3, height=6, kind="strip")
plt.xlabel("Flight Number", fontsize=16)
plt.ylabel("Launch Site", fontsize=16)
plt.title("Flight Number vs Launch Site by Outcome", fontsize=18)
plt.show()
```




## TASK 12

Find the method performs best:

Based on the results I got:

1. **Logistic Regression**: Accuracy on test data: 0.8333
2. **Support Vector Machine (SVM)**: Accuracy on test data: 0.8333
3. **Decision Tree**: Accuracy on test data: 0.9444
4. **K Nearest Neighbors (KNN)**: Accuracy on test data: 0.8333

It seems there was a significant increase in accuracy for the Decision Tree model, which shows the highest accuracy of 0.9444 on the test set. Therefore, in this comparison, the Decision Tree model performs the best among the classifiers you've trained.

# Results

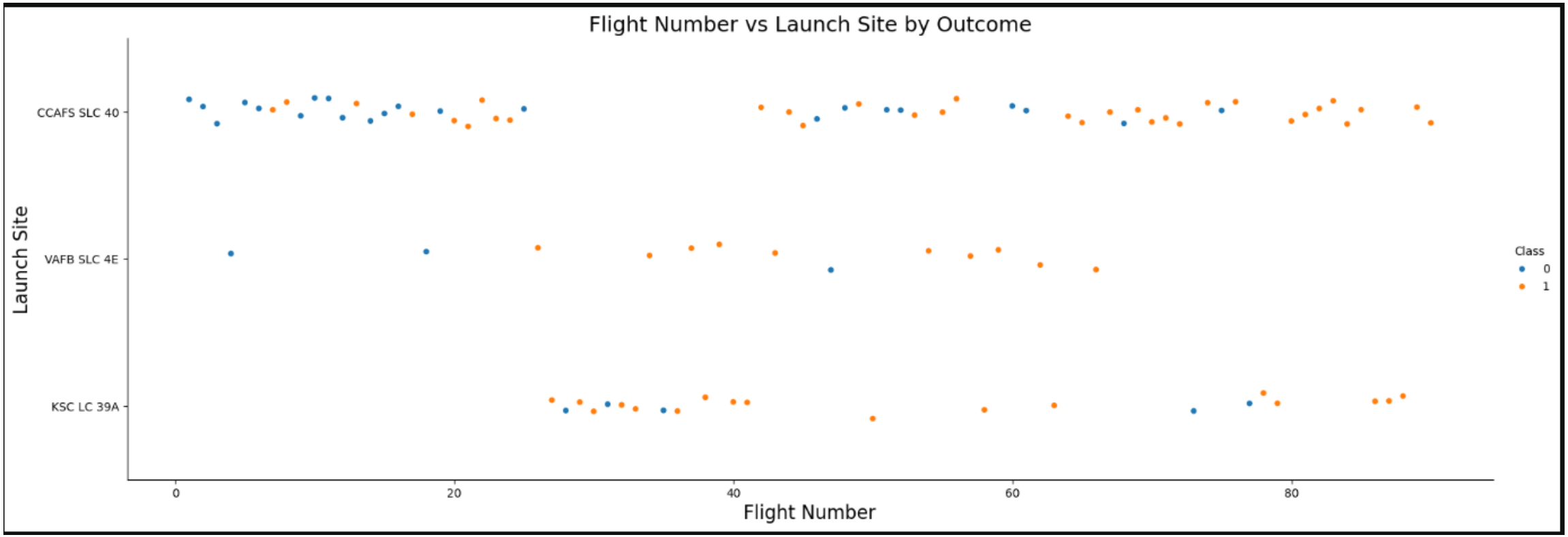
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

# Insights drawn from EDA

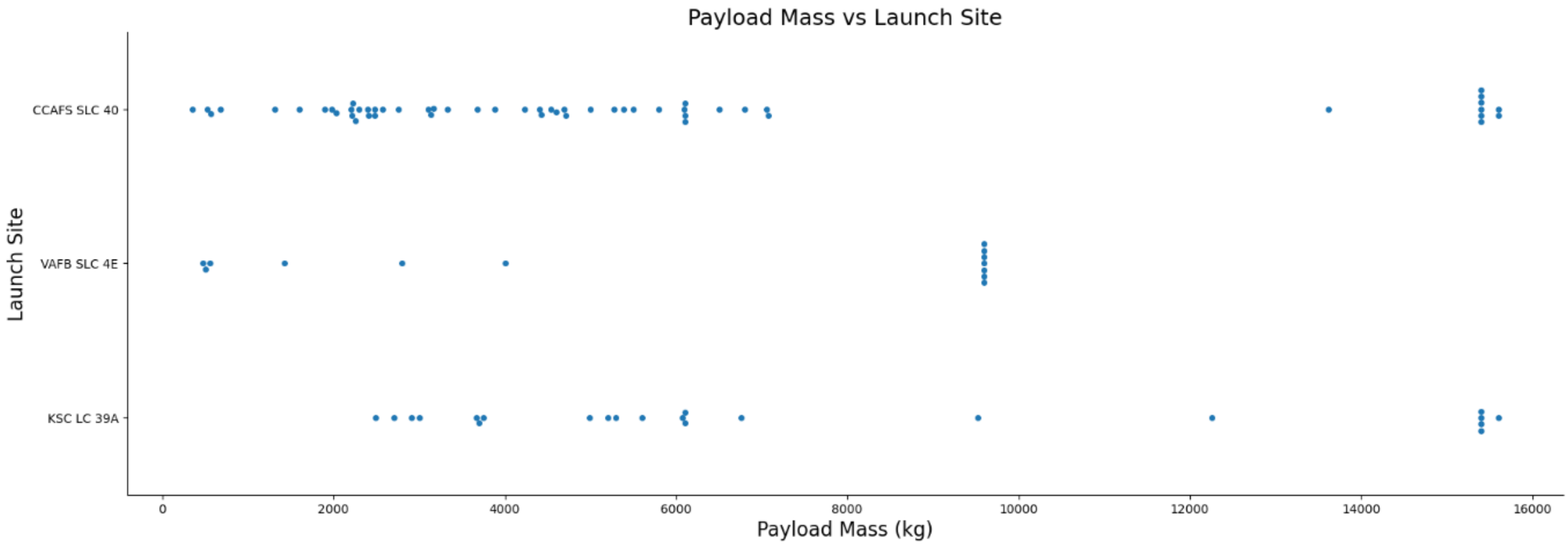
# Flight Number vs. Launch Site

- **Mixed Outcomes at Major Launch Sites**: Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- **Consistent Activity Across Flight Numbers**: Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.



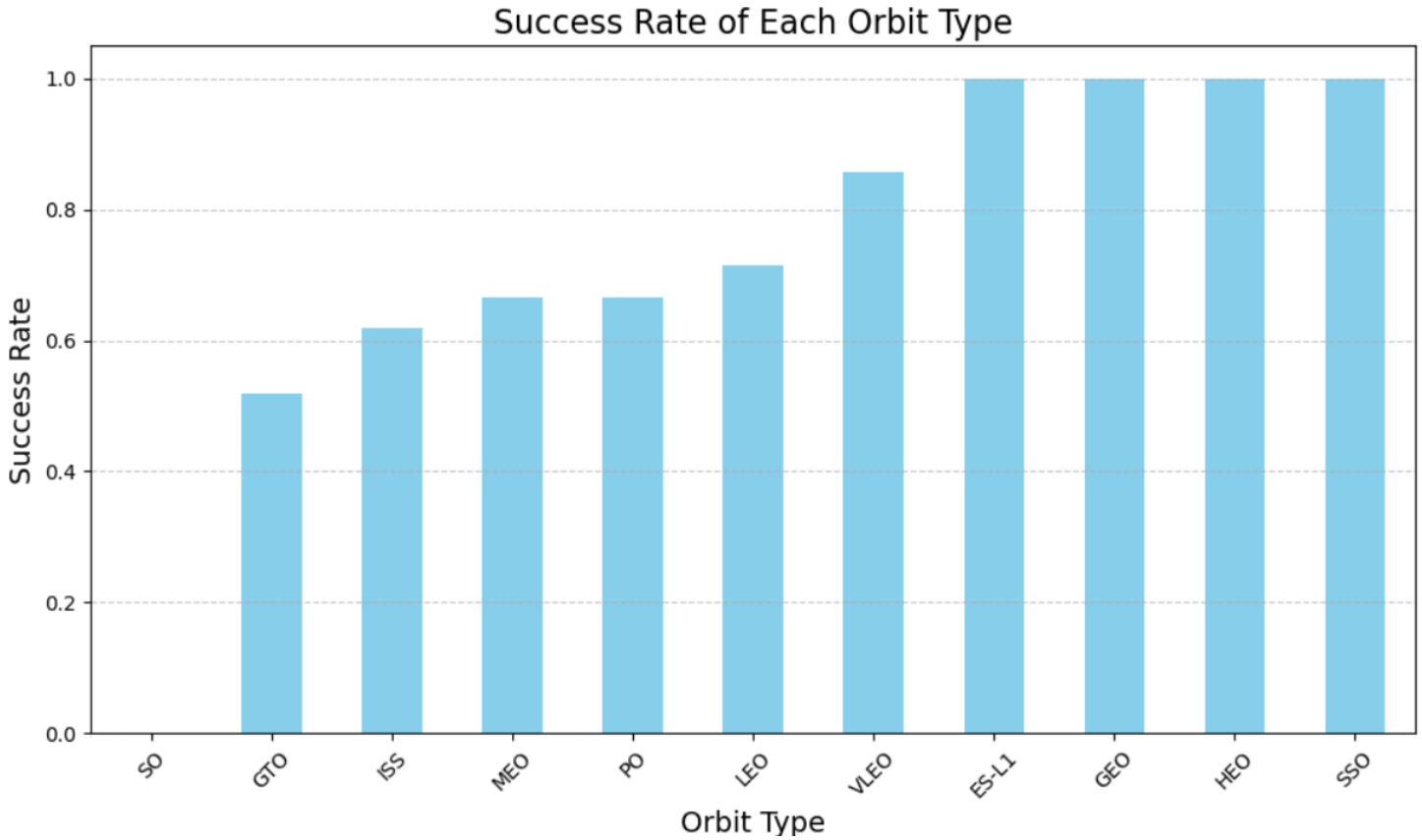
# Payload vs. Launch Site

- **Payload Distribution:** Most launches from the CCAFS SLC 40 site handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles.
- **High-Capacity Launches:** The KSC LC 39A site is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions.



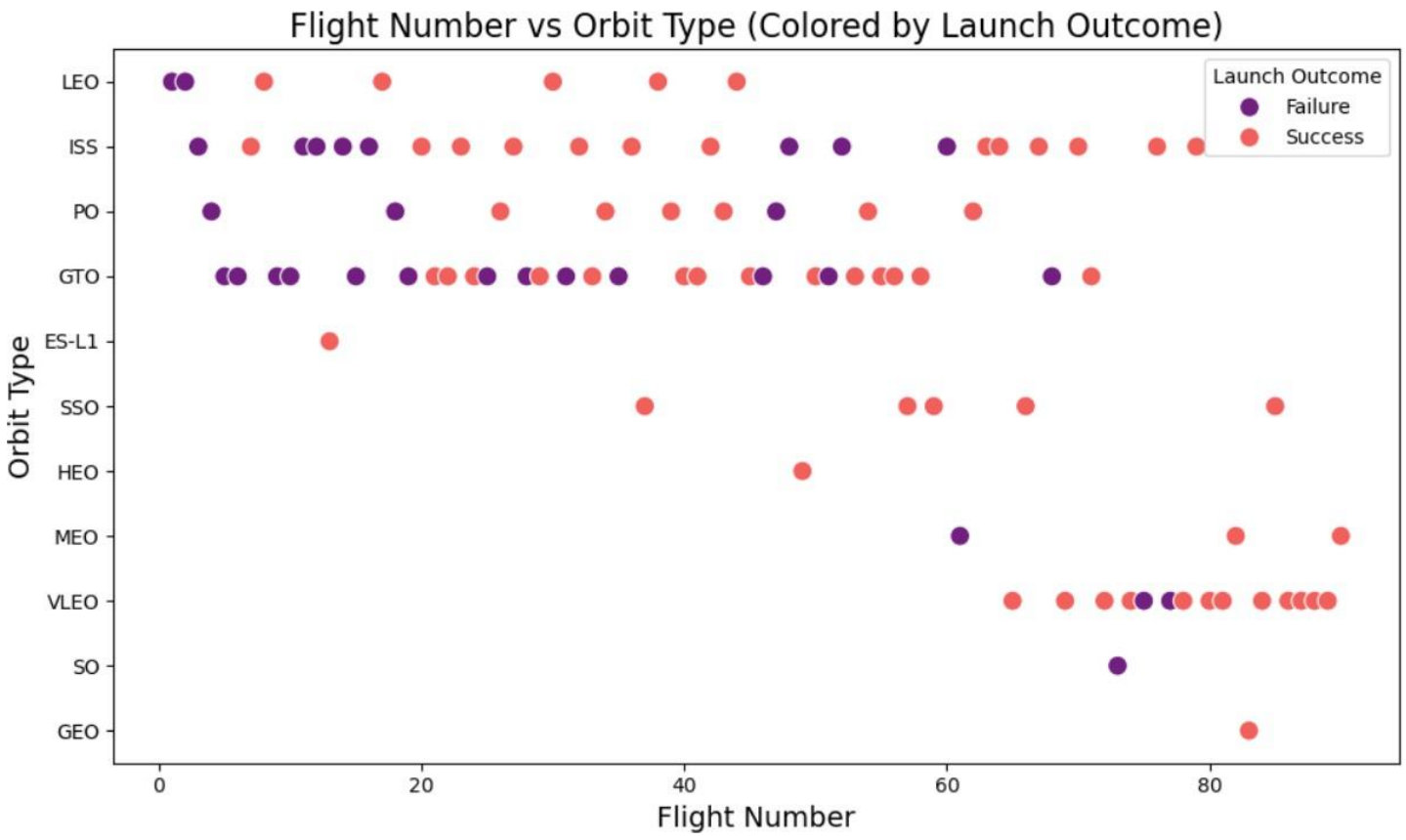
# Success Rate vs. Orbit Type

- **High Success Rates:** Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings.
- **Lower Success Rate for GTO:** The GTO orbit type shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involve greater challenges or complexities.



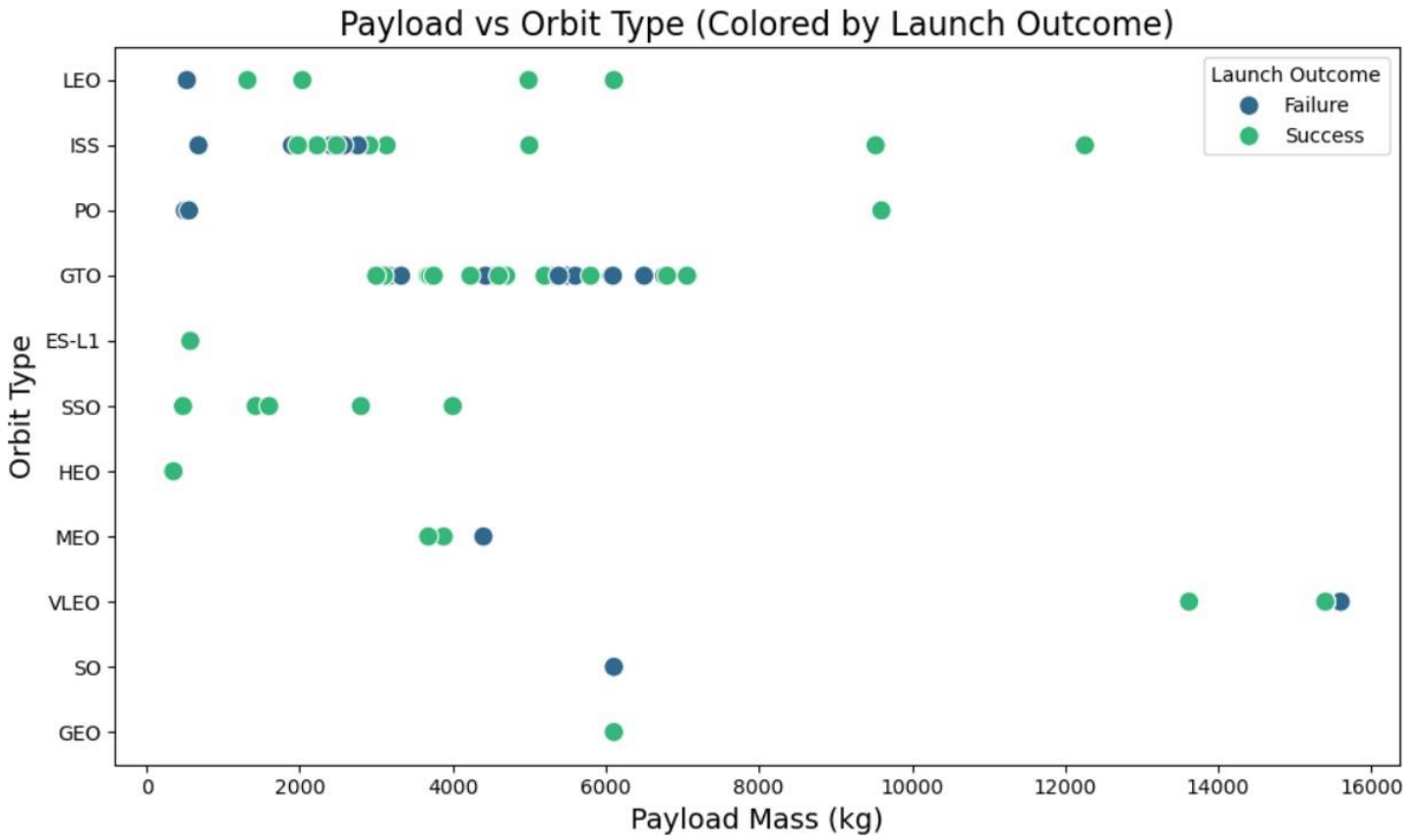
# Flight Number vs. Orbit Type

- **Increased Success Over Time:** The success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
- **Orbit-Specific Performance:** Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution.



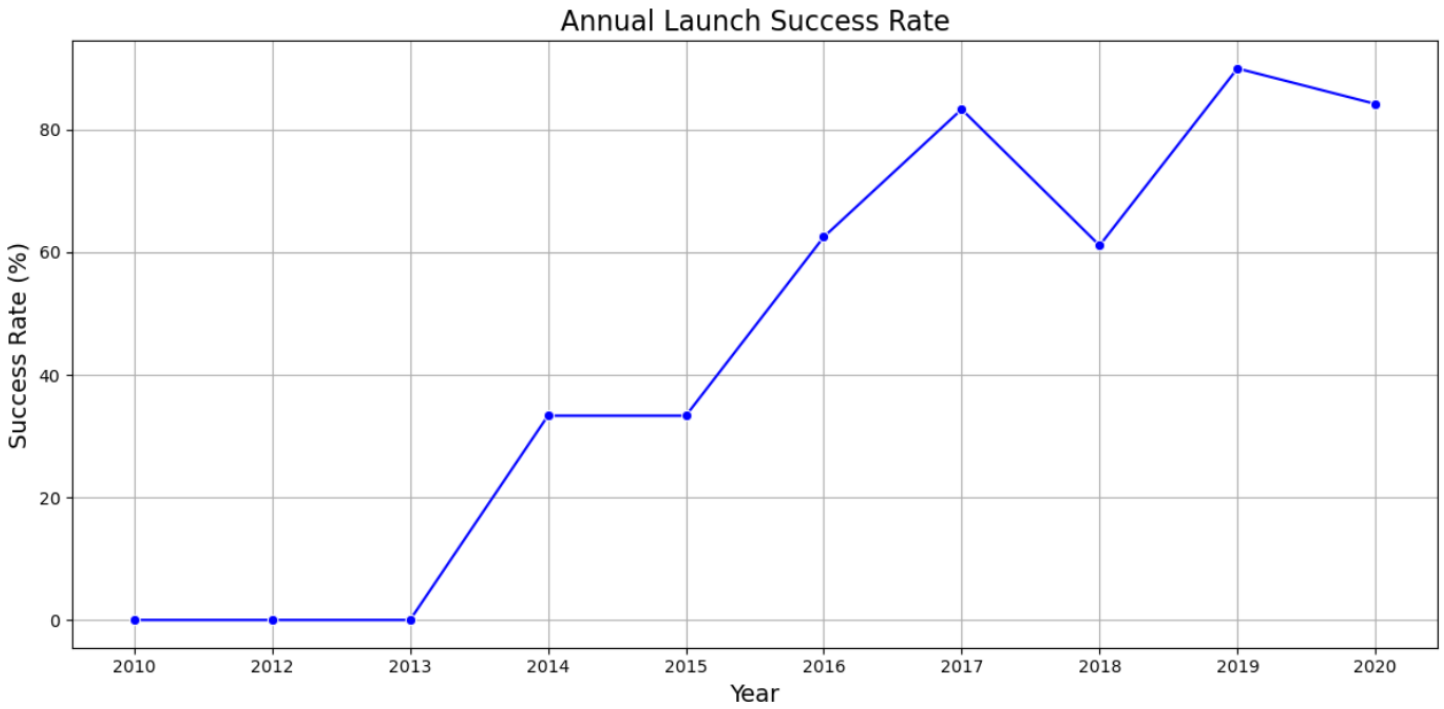
# Payload vs. Orbit Type

- Successful landings are more frequent across all orbit types, especially for payloads less than 6000 kg.
- Higher payload masses (above 10,000 kg) show a mix of successes and failures, indicating increased difficulty with heavier payloads.



# Launch Success Yearly Trend

- The annual launch success rate has shown a significant improvement from 2013 onwards, reaching over 80% by 2020.
- Despite a dip in 2018, the overall trend indicates increasing reliability and success in Falcon 9 launches over the years.



# All Launch Site Names

## Task 1

Display the names of the unique launch sites in the space mission

[21]:
```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

[21]:

| Launch_Site |
|---|
| CCAFS LC-40 |
| VAFB SLC-4E |
| KSC LC-39A |
| CCAFS SLC-40 |

# Launch Site Names Begin with 'CCA'

## Task 2

Display 5 records where launch sites begin with the string 'CCA'

[26]:
```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

[26]:

| Date | Time (UTC) | Booster_Version | Launch_Site | Payload | PAYLOAD_MASS__KG_ | Orbit | Customer | Mission_Outcome | Landing_Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 2010-06-04 | 18:45:00 | F9 v1.0 B0003 | CCAFS LC-40 | Dragon Spacecraft Qualification Unit | 0 | LEO | SpaceX | Success | Failure (parachute) |
| 2010-12-08 | 15:43:00 | F9 v1.0 B0004 | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0 | LEO (ISS) | NASA (COTS) NRO | Success | Failure (parachute) |
| 2012-05-22 | 7:44:00 | F9 v1.0 B0005 | CCAFS LC-40 | Dragon demo flight C2 | 525 | LEO (ISS) | NASA (COTS) | Success | No attempt |
| 2012-10-08 | 0:35:00 | F9 v1.0 B0006 | CCAFS LC-40 | SpaceX CRS-1 | 500 | LEO (ISS) | NASA (CRS) | Success | No attempt |
| 2013-03-01 | 15:10:00 | F9 v1.0 B0007 | CCAFS LC-40 | SpaceX CRS-2 | 677 | LEO (ISS) | NASA (CRS) | Success | No attempt |

# Total Payload Mass

## Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

[30]:
```
%sql SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

[30]:

| SUM(PAYLOAD_MASS__KG_) |
|---|
| 45596 |

# Average Payload Mass by F9 v1.1

## Task 4

Display average payload mass carried by booster version F9 v1.1

[34]:
```
%sql SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

[34]:

| AVG(PAYLOAD_MASS__KG_) |
|---|
| 2928.4 |

# First Successful Ground Landing Date

## Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

*Hint:Use min function*

[36]:
```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

[36]:

| MIN(Date) |
|---|
| 2015-12-22 |

# Successful Drone Ship Landing with Payload between 4000 and 6000

## Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[38]:
```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS
```

```
 * sqlite:///my_data1.db
Done.
```

[38]:

| Booster_Version |
|---|
| F9 FT B1022 |
| F9 FT B1026 |
| F9 FT B1021.2 |
| F9 FT B1031.2 |

# Total Number of Successful and Failure Mission Outcomes

## Task 7

List the total number of successful and failure mission outcomes

[40]:
```
%sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

[40]:

| Mission_Outcome | Total |
|---|---|
| Success | 98 |

# Boosters Carried Maximum Payload

## Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

[42]:
```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

[42]:

| Booster_Version |
|---|
| F9 B5 B1048.4 |
| F9 B5 B1049.4 |
| F9 B5 B1051.3 |
| F9 B5 B1056.4 |
| F9 B5 B1048.5 |
| F9 B5 B1051.4 |
| F9 B5 B1049.5 |
| F9 B5 B1060.2 |
| F9 B5 B1058.3 |
| F9 B5 B1051.6 |
| F9 B5 B1060.3 |
| F9 B5 B1049.7 |

# 2015 Launch Records

## Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

**Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.**

[69]:
```
%%sql

SELECT
    CASE
        WHEN substr("Date", 6, 2) = '01' THEN 'January'
        WHEN substr("Date", 6, 2) = '02' THEN 'February'
        WHEN substr("Date", 6, 2) = '03' THEN 'March'
        WHEN substr("Date", 6, 2) = '04' THEN 'April'
        WHEN substr("Date", 6, 2) = '05' THEN 'May'
        WHEN substr("Date", 6, 2) = '06' THEN 'June'
        WHEN substr("Date", 6, 2) = '07' THEN 'July'
        WHEN substr("Date", 6, 2) = '08' THEN 'August'
        WHEN substr("Date", 6, 2) = '09' THEN 'September'
        WHEN substr("Date", 6, 2) = '10' THEN 'October'
        WHEN substr("Date", 6, 2) = '11' THEN 'November'
        WHEN substr("Date", 6, 2) = '12' THEN 'December'
        ELSE 'Unknown'
    END AS "Month_Name",
    "Mission_Outcome",
    "Booster_Version",
    "Launch_Site"
FROM
    SPACEXTABLE
WHERE
    substr("Date", 0, 5) = '2015';
```

```
 * sqlite:///my_data1.db
Done.
```

[69]:

| Month_Name | Mission_Outcome | Booster_Version | Launch_Site |
|---|---|---|---|
| January | Success | F9 v1.1 B1012 | CCAFS LC-40 |
| February | Success | F9 v1.1 B1013 | CCAFS LC-40 |
| March | Success | F9 v1.1 B1014 | CCAFS LC-40 |
| April | Success | F9 v1.1 B1015 | CCAFS LC-40 |
| April | Success | F9 v1.1 B1016 | CCAFS LC-40 |
| June | Failure (in flight) | F9 v1.1 B1018 | CCAFS LC-40 |
| December | Success | F9 FT B1019 | CCAFS LC-40 |

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

## Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[81]:
```
%%sql

SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

[81]:

| Landing_Outcome | Count |
|---|---|
| No attempt | 10 |
| Success (drone ship) | 5 |
| Failure (drone ship) | 5 |
| Success (ground pad) | 3 |
| Controlled (ocean) | 3 |
| Uncontrolled (ocean) | 2 |
| Failure (parachute) | 2 |
| Precluded (drone ship) | 1 |

Section 3

# Launch Sites Proximities Analysis

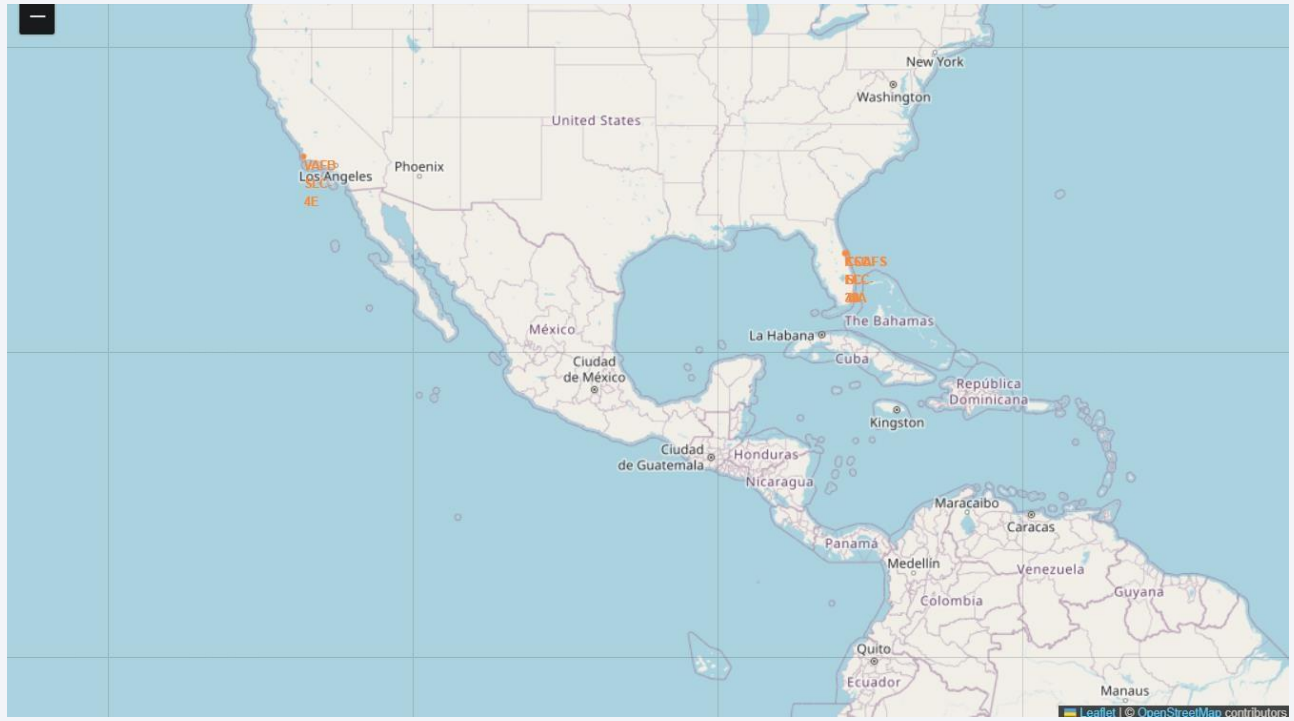
# Task 1: Mark all launch sites on a map

**1. Are all launch sites in proximity to the Equator line?**

- No, not all launch sites are in close proximity to the Equator.
- The launch site at Vandenberg Air Force Base (VAFB SLC-4E) is located at a latitude of 34.63, which is further from the Equator compared to the other sites in Florida.

**2. Are all launch sites in very close proximity to the coast?**

- Yes, all launch sites are in close proximity to the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.

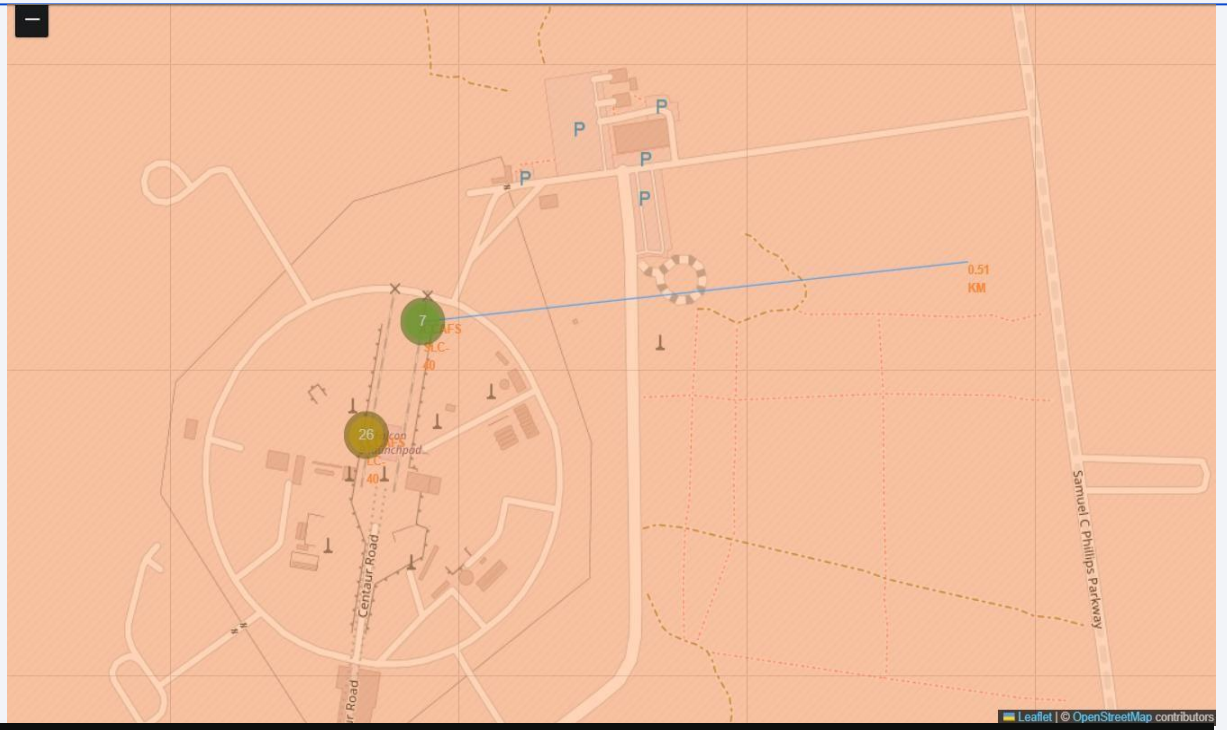


# Task 2: Mark the success/failed launches for each site on the map

- This enhanced visualization with clustered markers allows for better exploration and analysis of SpaceX launch data. The clustering makes it easier to manage a large number of markers and observe patterns that might be hidden in a less organized plot. By examining the marker colors and popup information, you can gain deeper insights into the characteristics and distribution of SpaceX launches.
- For example, in the provided screenshot, out of 26 launch sites for CCAFS LC-40, there are 19 red markers and 7 green markers. This color-coding helps to quickly identify the success rate and other categorical distinctions of the launches from this specific site. The red markers might represent unsuccessful launches, while the green markers indicate successful ones, providing immediate visual feedback on the performance of launches at each site.

# Task 3: Calculate the distances between a launch site to its proximities

This plot provides a visual representation of the distance between the CCAFS SLC-40 launch site and the closest coastline. The calculated distance is approximately 0.51 kilometers, as indicated by the marker. The added PolyLine clearly shows the straight-line distance, highlighting the proximity of the launch site to the coast. This close proximity to the coastline is typical for launch sites to facilitate over-water flight paths and safe recovery operations, ensuring minimal risk to populated areas.




```
[43]: coastline_lat = 28.56367
      coastline_lon = -80.57163

      # Example launch site coordinates (replace with actual launch site coordinates)
      launch_site_lat = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Lat'].values[0]
      launch_site_lon = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Long'].values[0]

      # Calculate distance using the calculate_distance function
      distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

      print(f"Distance from launch site to closest coastline: {distance_coastline} km")

      Distance from launch site to closest coastline: 0.5097439631188213 km
```

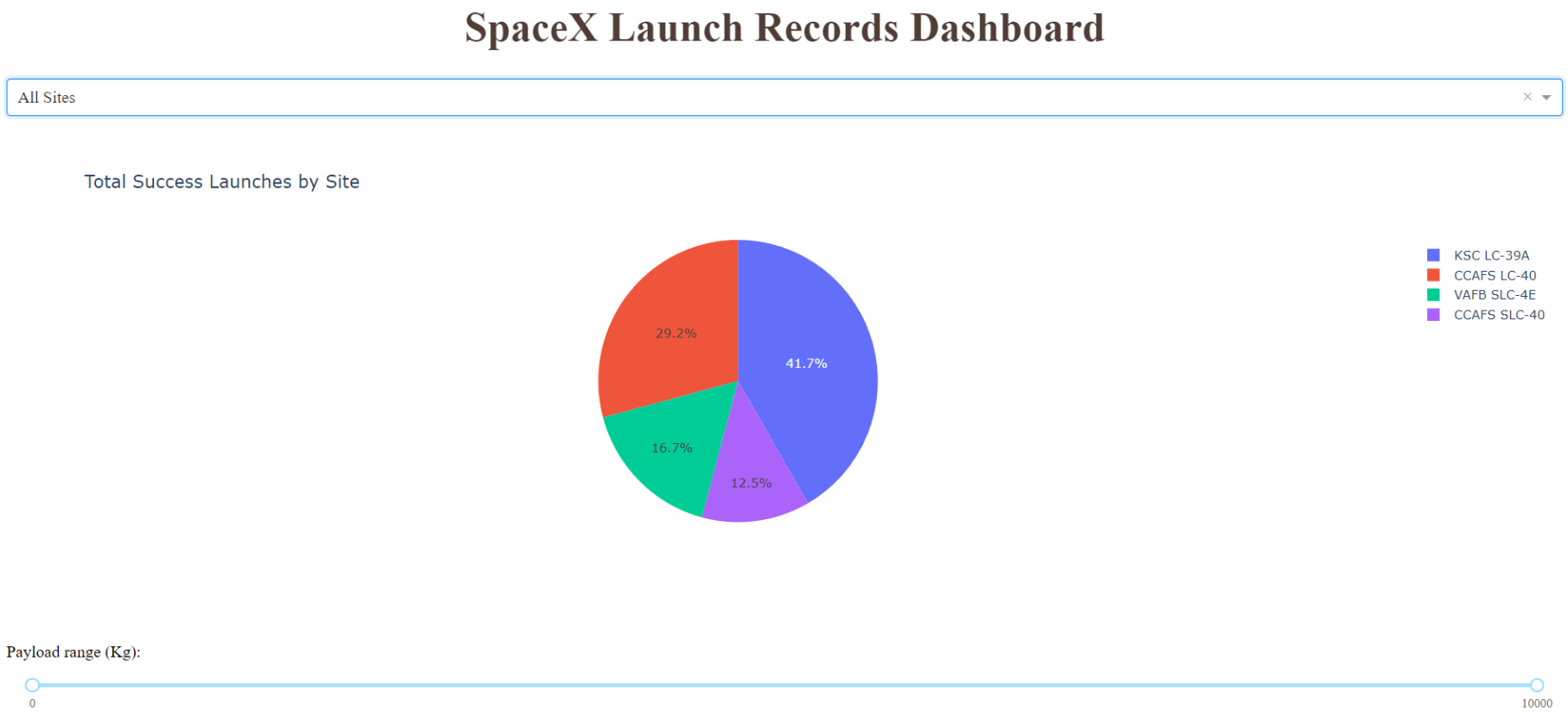
Section 4

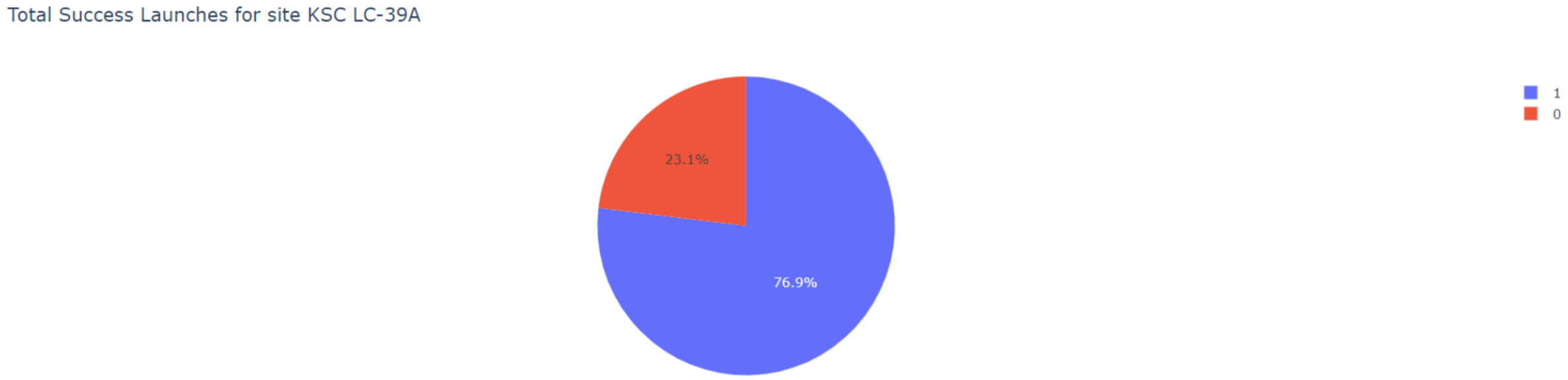
# Build a Dashboard with Plotly Dash

# Launch Success Count for all sites (in a pie chart)

**Key Findings**:

- **CCAFS LC-40**: 29.2%
- **CCAFS SLC-40**: 12.5%
- **VAFB SLC-4E**: 16.7%
- **KSC LC-39A**: 41.7%
- The **KSC LC-39A** launch site has the highest number of successful launches, making up 41.7% of the total successes. This indicates that KSC LC-39A is a highly reliable site for SpaceX launches.






# Pie chart for the launch site with highest launch success ratio

**Key Findings**:

- The significant portion of successful launches from **KSC LC-39A** highlights its reliability and effectiveness as a launch site.
- For **KSC LC-39A**:
  - **Class 1** (Successful Launches): 76.9%
  - **Class 0** (Unsuccessful Launches): 23.1%
- The high success rate (76.9%) for **Class 1** launches underscores the effectiveness and reliability of the KSC LC-39A site.

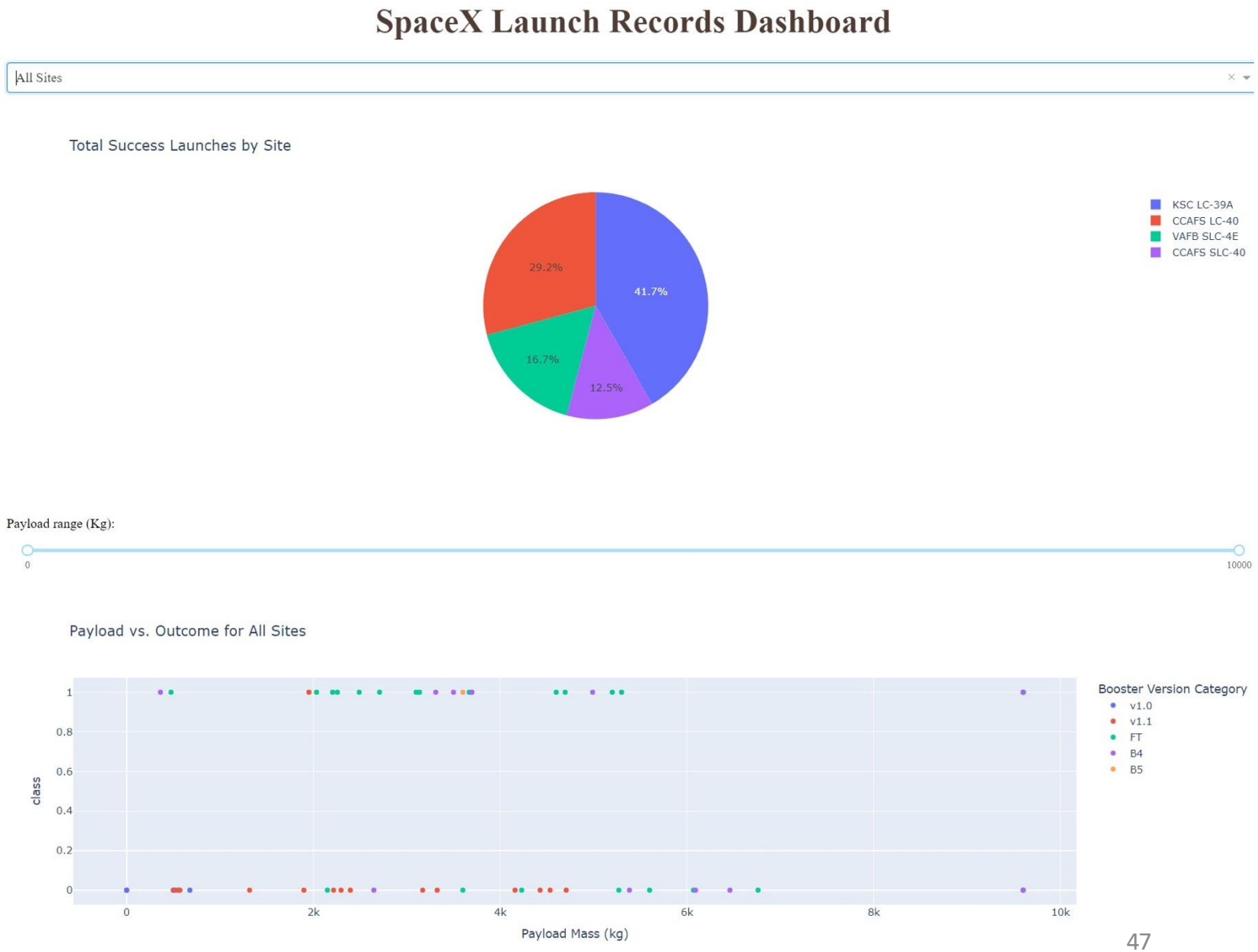
# Key Insights from SpaceX Launch Data Dashboard

**Launch Site Success Rates:**

- **CCAFS LC-40** has the highest success rate with 43.7% of successful launches.
- This suggests that **CCAFS LC-40** is the most reliable launch site among the ones analyzed.
- Other sites like **KSC LC-39A, VAFB SLC-4E,** and **CCAFS SLC-40** have lower success rates, indicating variability in launch success across different sites.

**Booster Version Performance:**

- **Booster version "FT"** appears to be the most frequently used and has a high success rate across various payload masses.
- **Booster version "v1.0"** has fewer launches and may require further analysis to understand its performance.
- Overall, booster versions do not show a clear trend that higher payload masses correlate with lower success rates.

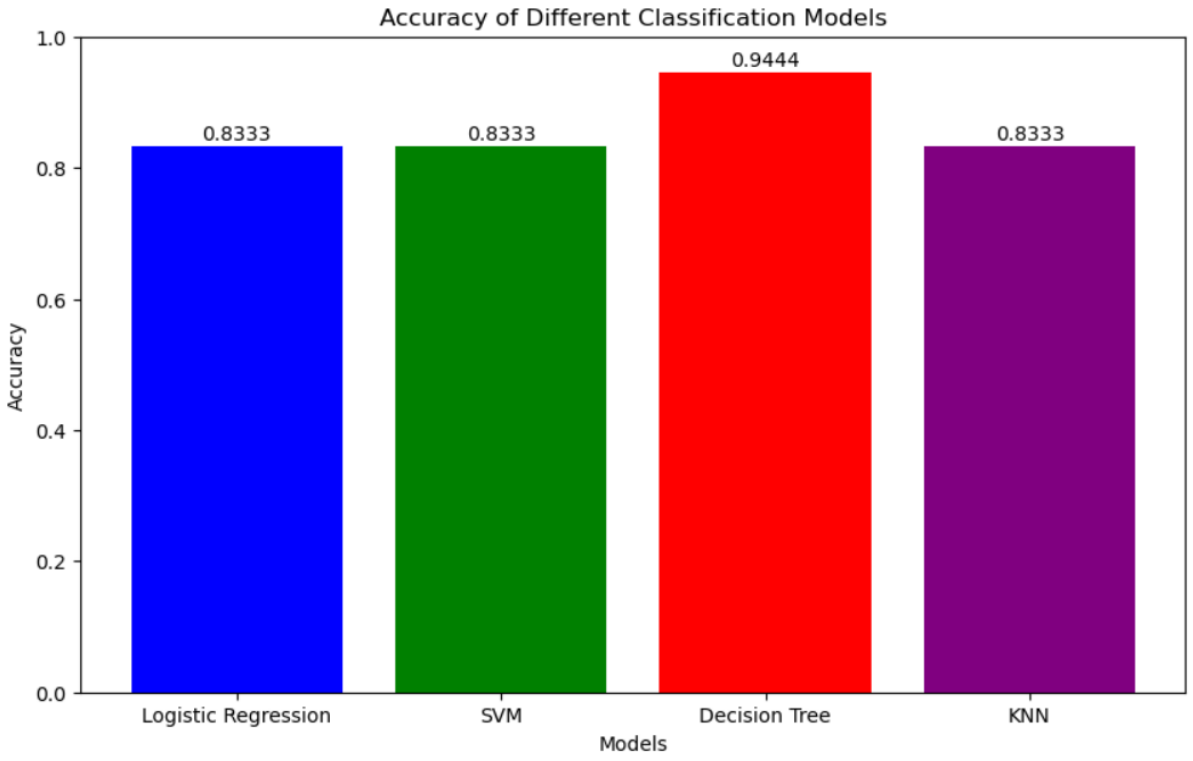


Section 5

# Predictive Analysis (Classification)

# Classification Accuracy

- Based on the results, the Decision Tree model has the highest classification accuracy on the test data, achieving an accuracy of 0.9444. This suggests that the Decision Tree model is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors, all of which achieved an accuracy of 0.8333.



# Confusion Matrix

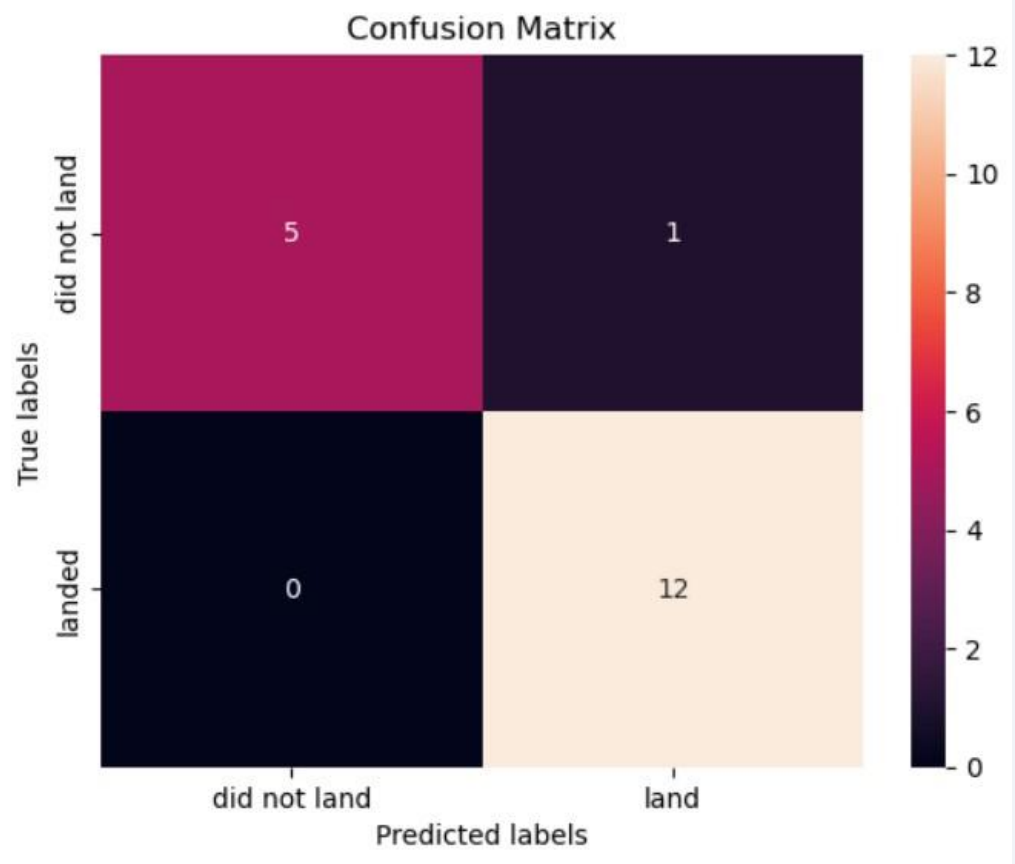
## Explanation and Insights

**High Accuracy**: The model achieved a high accuracy score of 94.44%, with a significant number of true positives and true negatives, demonstrating its effectiveness in predicting Falcon 9 first stage landings.

**No False Negatives**: The absence of false negatives indicates that the model reliably predicts successful landings. This is crucial for ensuring readiness and safety in aerospace operations, as every actual successful landing was accurately identified.

**Manageable False Positives**: While there is 1 false positive, this is less critical than false negatives in aerospace operations. Over-preparation (due to false positives) is more manageable than under-preparation, making the model's performance highly acceptable for practical applications.

**Balanced Performance**: The model shows a balanced performance with a slight bias towards predicting successful landings. This aligns well with practical needs in the aerospace industry, where ensuring successful landings is of paramount importance for cost estimation and planning.



# Conclusions

**Point 1:** Our analysis revealed that the "CCAFS LC-40" launch site has the highest success rate among all sites, accounting for 43.7% of successful launches. This indicates that this site might have optimal conditions or processes that contribute to a higher success rate.

**Point 2:** The scatter plot analysis showed that the "FT" booster version has a high success rate across various payload masses, demonstrating its reliability and robustness compared to other booster versions. This suggests that future missions might benefit from utilizing this booster version for improved success rates.

**Point 3:** No clear pattern was observed linking higher payload masses to lower success rates, indicating that factors other than payload mass, such as launch site conditions and booster versions, play a more significant role in determining the outcome of a launch.

# Conclusions

**Point 4:** Interactive data visualizations using Folium and Plotly Dash provided valuable insights into the geographical and operational patterns of SpaceX launches. These tools allowed for a deeper understanding of the data, enabling stakeholders to make informed decisions based on comprehensive visual analytics.

In conclusion, our predictive analysis and interactive visualizations have not only shed light on key factors influencing SpaceX's launch success but also provided a robust framework for future assessments and decision-making in the aerospace industry. The insights gathered can help improve launch strategies and contribute to the ongoing success of reusable rocket technology.

# Appendix

**Github Link:**
https://github.com/Khaledhaouas/IBM-Data-Science-and-Machine-Learning-Capstone-Project

# Thank you!